

Power-Share Setpoint Sweep under Trapezoidal Load: Bench Results and a Minority-Channel Dropout Limit Cycle

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Abstract

Seven bench runs (TP0007–TP0013) sweep the power-share setpoint of the DC balancer board’s droop-based share controller across the full $[0, 1]$ range under an identical trapezoidal motor-current profile. Steady-state tracking is excellent and unbiased at every setpoint (hold-window mean error $< 10^{-3}$ across the sweep), and interior setpoints settle in under 2 s with no actuator saturation. Two runs, however, expose a previously unobserved failure mode: at asymmetric setpoints (0.30 and 0.85) and low total current ($\lesssim 1.2$ A), the loop enters a sustained 17–18.5 Hz *minority-channel dropout limit cycle*. In its severe form (TP0010, setpoint 0.30) the two boost channels conduct mutually exclusively, reconnect spikes reach 3.6 A, and the bus collapses below 15 V sixty-four times — to a minimum of 6.55 V on a 15.9 V rail — without raising any fault. This paper documents the sweep, characterizes the limit cycle, evaluates candidate mechanisms, and recommends firmware and test-procedure changes. The event is logged as a nondestructive datapoint in the board’s bring-up failure record.

1 Test setup and data provenance

All runs were performed on the 20260622 balancer board in State-98 test mode using the trapezoidal current profile T 6 3 1: motor current command ramps $0 \rightarrow 6$ A at 1 A/s, holds 6 A for 3 s, then ramps back to zero (≈ 15 s total). The Youla- H_∞ power-share controller (`share_controller.h`, $K_{\text{DROOP}} = 0.300 \Omega$) is active throughout; the swept variable is the constant share setpoint r^* (fuel-cell fraction of total channel current), one value per run. Firmware is the pre-versioning build (“fw 0” in the version ledger) compiled with `BENCH.TEST=1`, which arms *overvoltage faults only* — a fact that matters in Section 3. The bus regulates at ≈ 15.9 V no-load.

Data come from the on-board SD bench logger (1 kHz nominal; measured ≈ 865 Hz effective, 1.15 ms median sample interval). All seven captures are complete: zero dropped records, zero missed periods, clean trailer close. Decoding and the per-run overview figures were produced by the `tools/benchlog_analysis` pipeline; the zoomed and summary figures in this paper are generated by the accompanying `make_figs.py`. No oscilloscope was armed during the sweep, so all quantities are ADC-path readings through the firmware’s averaging chain (current LSB ≈ 8.06 mA); sub-millisecond structure is not observable.

Table 1: The sweep, in session order. Hold-window statistics are computed over the 3 s 6 A plateau (`trap_phase=1`). “Settle” is time from load onset (filtered total current > 0.15 A) to $|e| < 0.05$ sustained 0.2 s; 0.0 means the loop was already within band at onset.

Run	r^*	Mean err	σ	Settle [s]	$\bar{I}_{\text{tot}}$ [A]	Anomaly
TP0007	0.50	−0.00002	0.010	0.0	1.20	none (baseline)
TP0008	0.70	−0.00000	0.011	0.0	1.38	none
TP0009	1.00	−0.00000	0.0001	0.0	1.55	degenerate: BT off, MDACs railed
TP0010	0.30	−0.0044	0.108	1.81	1.50	17 Hz limit cycle; bus to 6.5 V
TP0011	0.00	+0.00000	0.0001	0.0	1.61	degenerate: FC off, loop frozen
TP0012	0.15	−0.00000	0.010	0.0	1.65	g_{FC} within 5 % of rail
TP0013	0.85	+0.00002	0.043	1.89	1.72	18.5 Hz BT dropout chatter

2 Steady-state tracking: unbiased across the full span

Figure 1 summarizes the sweep. Hold-window mean error is below 10^{-3} at every setpoint from 0 to 1 — the integral action carries no bias anywhere in the span, matching the design intent of the Youla controller’s exact $T(0) = 1$ constraint. Interior setpoints (0.3–0.7) show $\sigma \approx 0.01$ with the droop-gain commands comfortably mid-band, and Figure 2 shows the representative clean run at $r^* = 0.5$. The endpoints are degenerate by construction: at $r^* = 1$ the BT channel simply stops conducting and both MDAC commands sit pinned at the band edges (an open loop with zero error); at $r^* = 0$ the loop freezes mid-range with the FC channel dark. Neither endpoint misbehaves.

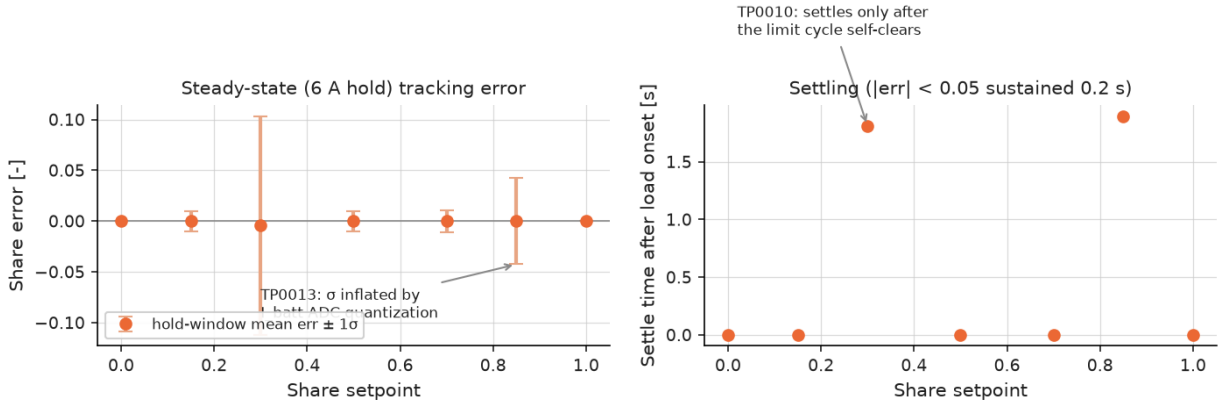


Figure 1: Sweep summary. Left: hold-window tracking error is unbiased at every setpoint; the two inflated σ bars belong to the limit-cycle runs (TP0010) and the quantization-corrupted high-share run (TP0013). Right: settling time; interior and endpoint runs are within band at load onset, while the two anomalous runs take ≈ 1.8 – 1.9 s.

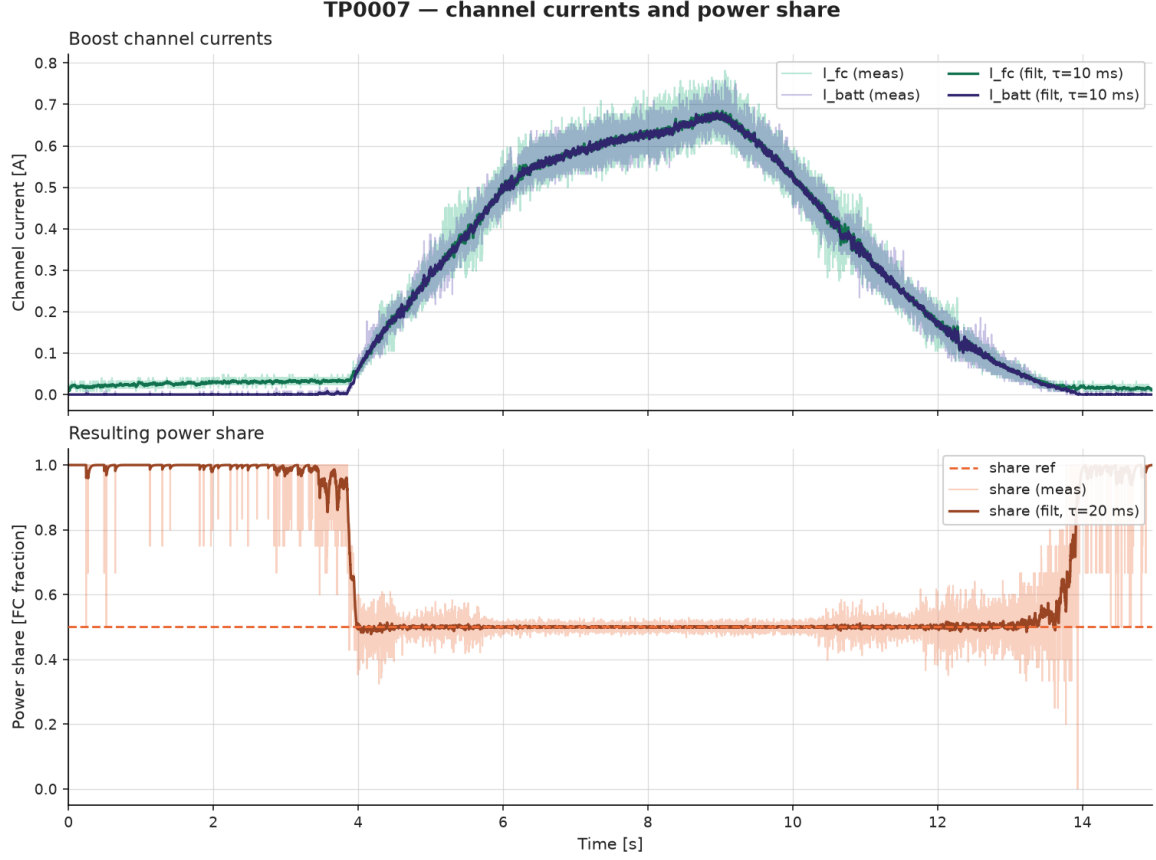


Figure 2: Representative clean run (TP0007, $r^* = 0.5$): symmetric channel currents through the full trapezoid, share locked to the setpoint whenever total current is above the measurement floor. The share excursions before load onset and after load release are the known low-current region where the share ratio is numerically undefined, not a control failure.

Actuator headroom, however, is *asymmetric*. At $r^* = 0.15$ the FC droop command rides 0.85–0.91 (within $\approx 5\%$ of its ceiling) yet the run is the fastest-settling of the sweep. The mirror case $r^* = 0.85$ needs $g_{BT} \approx 0.90$ – 0.99 and g_{FC} pinned at its 0.175 floor for 85% of the loaded window — and misbehaves (Section 3.2). The usable share band is therefore not symmetric about 0.5: the FC-heavy side runs out of authority first. This should inform the pending bench calibration of K_{DROOP} and the droop-ratio band rather than being read as a controller defect.

3 The minority-channel dropout limit cycle

3.1 Severe form: TP0010, $r^* = 0.30$

From $t = 4.33\text{s}$ to 6.15s — mid ramp-up, with the motor command between 4.3 and 6 A but total channel current still below $\approx 1.2\text{ A}$ — the loop enters a sustained relaxation oscillation with a median period of 58–59 ms ($\approx 17\text{ Hz}$). Figure 3 shows the full window; Figure 4 resolves individual cycles.

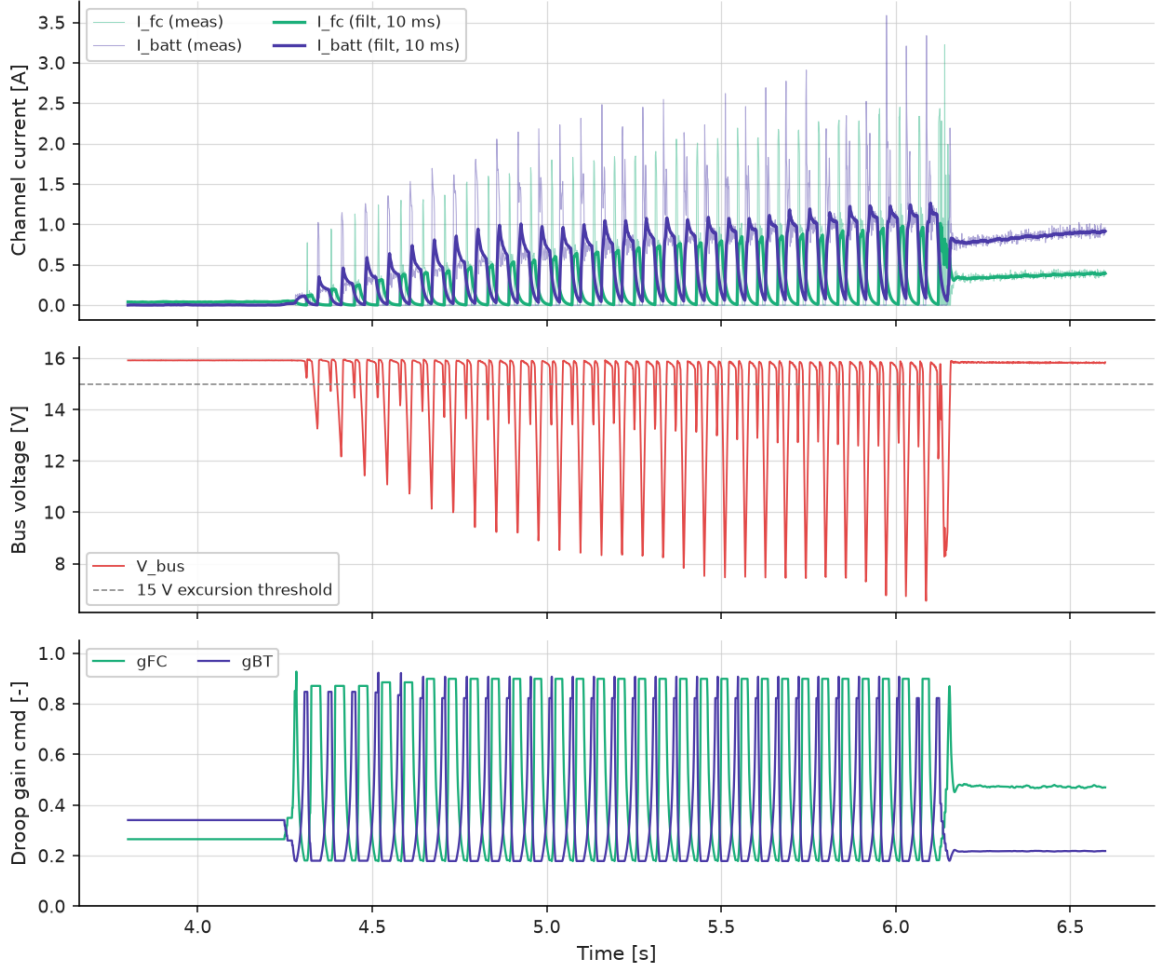


Figure 3: TP0010 limit-cycle window. Top: channel currents alternate with reconnect spikes growing to 3.6 A as the load ramp deepens. Middle: 64 bus excursions below 15 V, reaching 6.55 V; sag depth grows monotonically with the commanded load. Bottom: the droop MDAC commands bang between their extremes in antiphase. The cycle self-clears at $t = 6.15$ s once total current exceeds ≈ 1.2 A, and the remainder of the run is clean.

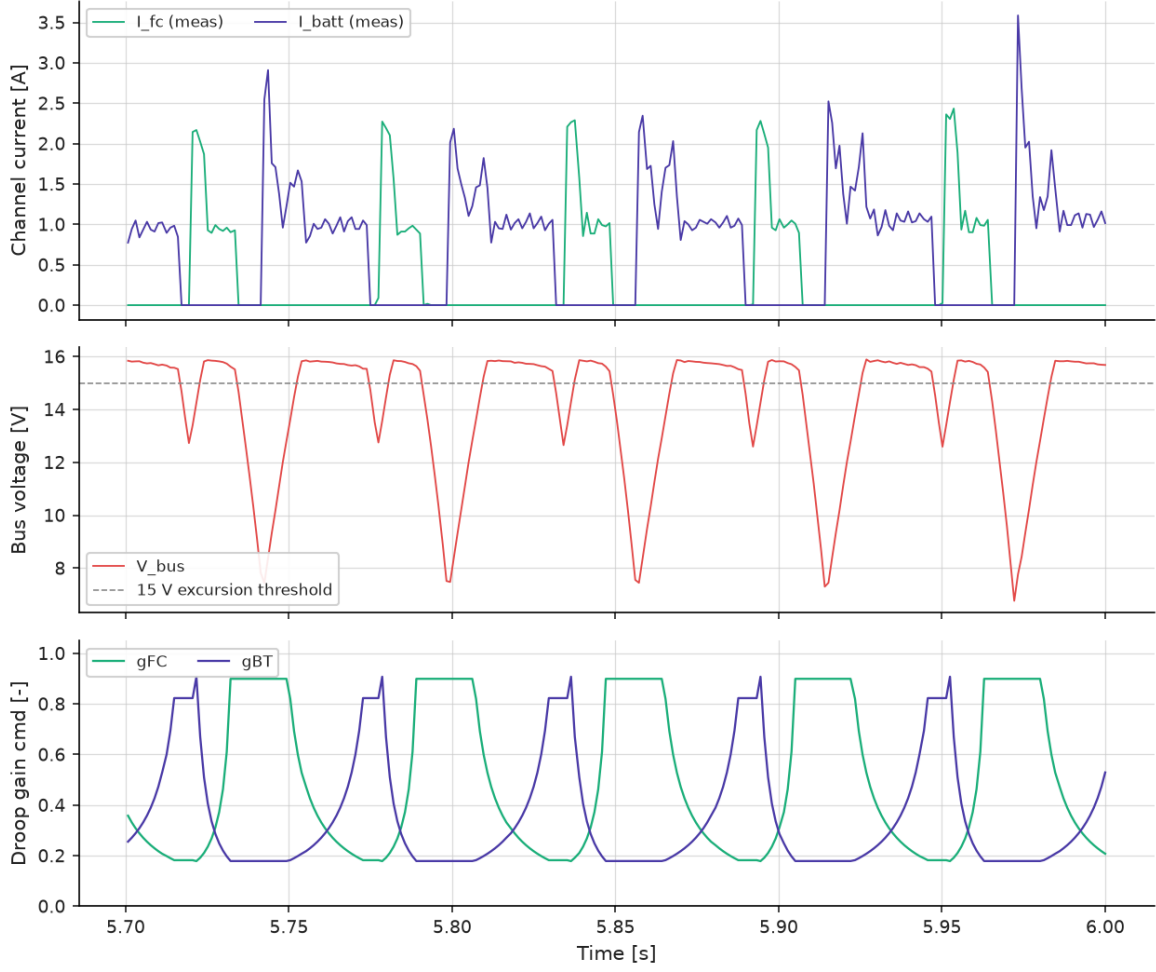


Figure 4: Five consecutive cycles of the TP0010 limit cycle. The two channels conduct *mutually exclusively* (both conducting in only 0.1 % of samples); each handoff includes a 10–18 ms window in which *both* channels read zero current and the bus — carrying only its $\approx 40 \mu\text{F}$ of ceramics — falls as low as 6.5 V before the incoming channel reconnects with a 2–3.6 A spike.

Three features deserve emphasis. First, the collapse is real and repeated: 64 excursions below 15 V in 1.8 s, with sag depth scaling with the commanded load (13.3 V at $I_{\text{cmd}} = 4.4 \text{ A}$ down to 6.5 V at 6 A). Second, *no fault was raised*: the `BENCH.TEST` build arms overvoltage checks only, and every excursion is downward — the event is invisible to `detectFaults()` as currently configured on the bench. Third, the cycle is self-limiting in this run: once the trapezoid pushes total current past $\approx 1.2 \text{ A}$ the loop converges and the remainder of the run is indistinguishable from the clean runs ($\sigma = 0.011$, no saturation). A brief re-entry occurs on the low-current tail at $t \approx 13.5 \text{ s}$.

3.2 Mild form: TP0013, $r^* = 0.85$

The mirror-side asymmetric setpoint shows the same signature at 18.5–18.7 Hz, but as minority-channel (battery) dropout chatter without bus collapse (Figure 5): during both ramps the BT channel repeatedly falls to exactly zero current (pinned there for 0.77 s of the 1.5 s ramp-up band and 1.36 s of the 2.1 s ramp-down band), the measured share slams to 1, and g_{BT} rails before the channel snaps back. Bus disturbance is limited to $\approx 100 \text{ mV}$ notches. The run never meets the

strict settling criterion ($|e| < 0.03$ for 200 ms) anywhere; its mid-plateau is acceptable but has the worst steady-state spread of the sweep.

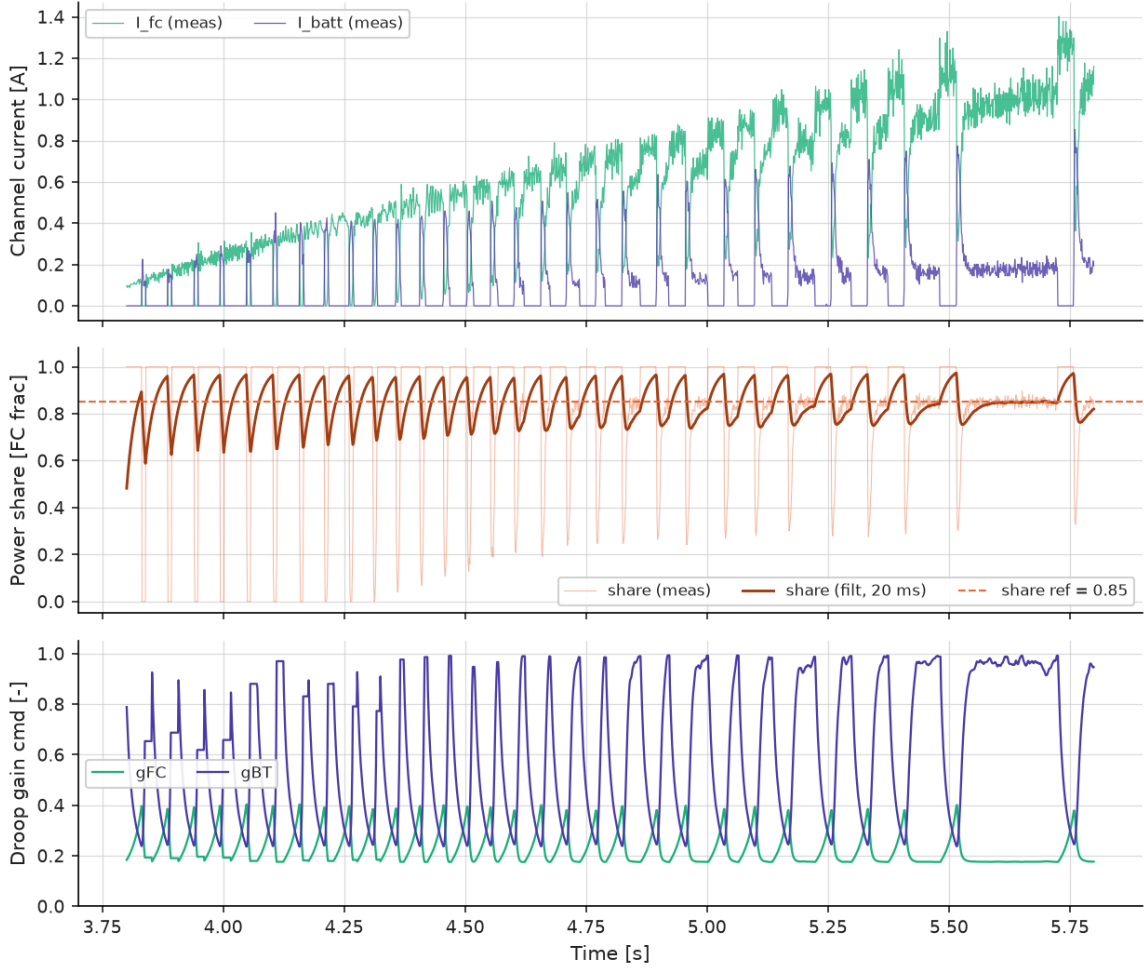


Figure 5: TP0013 ramp-up dropout band. The BT channel (violet) repeatedly drops out of conduction entirely; measured share (middle) saws between the setpoint and 1.0 at ≈ 18.5 Hz; g_{BT} (bottom) rails toward 1.0 on every dropout while g_{FC} sits pinned at its 0.175 floor. Note the events thin out as total current grows toward the plateau.

A separate measurement artifact compounds the picture at this setpoint: the commanded battery current ($\approx 15\%$ of ≈ 0.5 A ≈ 75 mA) is only ≈ 9 LSB of the 8.06 mA current-sense quantization, and I_{BT} quantizes to exactly zero for 23% of the loaded segment, pegging the computed share at exactly 1.0. Excluding those samples reveals a genuine -0.03 share bias at $r^* = 0.85$ that the raw mean coincidentally cancels. Setpoint 0.85 at these load levels is where the current-sense LSB begins to corrupt the share estimate itself.

3.3 Mechanism assessment

The data support a *starved-minority-channel dropout loop*: at low total current, the droop command needed to hold the minority channel at a small fraction pushes that channel out of conduction entirely (its RT1987 ideal diode blocks once the droop-raised source characteristic falls below the

bus); the measured share then slams to 0 or 1, the controller drives both MDACs toward their rails, the starved channel reconnects with a large spike, and the cycle repeats. This explains the frequency similarity between the two runs, the mutual exclusivity, the confinement to low total current, and the self-clearing once the minority channel's commanded current rises above its conduction floor.

What that loop alone does *not* explain is TP0010's 10–18 ms *both-channels-off* windows. Two candidates remain open, and the mechanism is explicitly **unconfirmed** pending a scope capture:

1. both boosts are simultaneously droop-commanded out of conduction during the crossover (a pure control-loop artifact of the antiphase rail-to-rail MDAC slew);
2. the RT1987 bus switches' short-circuit-protection cut/retry participates: the 58–59 ms period is suspiciously close to the 64 ms `tSCP_RST` retry fingerprint documented in the board's bring-up record (captures 9–10), and both bus switches have carried 100 nF soft-start capacitors since 2026-08-07, lengthening every reconnect.

A scope capture aimed at this question was taken on 2026-08-11 (capture 12 in the bring-up record; Figure 6): a repeat of the TP0010 condition with VBUS, V-MOT, and the battery-channel INA monitored, single-shot trigger on VBUS falling through 14 V. It *narrows* the question without closing it:

- **Confirmed: the events are total source-feed dropout windows.** VBUS and V-MOT sag *together* — slow linear ramps of $\approx 3\text{--}4\text{ V}$ over tens of milliseconds at a 67.6 ms cursor period (14.8 Hz), each ended by a sharp recovery synchronized with a $\approx 1\text{ A}$ reconnect burst on the battery INA. Because the motor switch conducts throughout, the bus rides the combined $\approx 1\text{--}1.5\text{ mF}$ motor+VESC capacitance during each window — which is why the decay is slow, and which independently corroborates the SD log's 10–18 ms both-channels-off reading.
- **Ruled out: the VBUS→V-MOT switch as a cutting element** (the two nodes track through every dip).
- **Still open: candidates (1) vs (2).** Both predict exactly this bus signature, and the 67.6 ms period is consistent with either (the RT1987 retry timer is 64 ms; the SD-log median was 58–59 ms). The sharper discriminator, if needed: probe a *boost output* alongside VBUS during a dropout — reconnect at the voltage crossing indicates droop-commanded blocking, reconnect on a fixed 64 ms cadence indicates SCP.

Practically, the discrimination no longer gates the fix: both surviving candidates are downstream of the droop loop railing both MDACs antiphase, so the mitigation set of Section 5 (feasibility-gated integrator hold, MDAC slew limiting) is the correct next step under either mechanism.

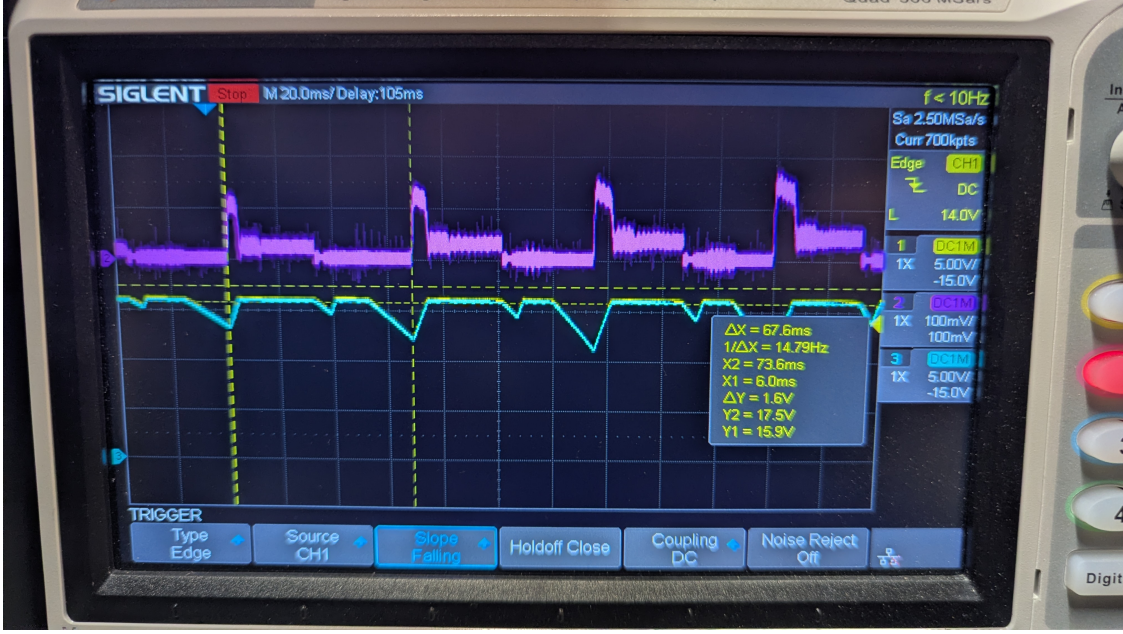


Figure 6: Capture 12 (2026-08-11): TP0010-condition repeat, 20 ms/div. CH1 VBUS (yellow, 5 V/div) and CH3 V-MOT (blue, 5 V/div) overlap and sag together in $\approx 3\text{--}4\text{ V}$ sawtooth ramps at a 67.6 ms period (cursors); CH2 battery INA (purple, 1 A/div) shows the dropout–reconnect–spike–conduction pattern synchronized with the voltage recoveries. The flat yellow line near 15.9 V is the Y1 measurement cursor, not the VBUS trace.

3.4 Why a nondestructive event matters

Each reconnect is a load-dump-class transient sourced by a boost converter, on a board whose failure history (five destroyed TPS61288 converters) traces to exactly this event class. The per-event energy here is far below the destructive episodes, and both boost converters carry the validated hot-loop bodge capacitors, which bounds the known overshoot mechanism — but TP0010 delivered ≈ 33 collapse/reconnect events in under two seconds, a duty cycle the power stage has not been characterized under. Capture 12 confirms the whole bus — sources to VESC — rides each dropout’s decay, so until the firmware mitigation lands, repeated runs at $r^* \approx 0.3$ with ramping load should be treated as a known stressor. The event is logged in `docs/boost-bringup-debug.md` with a corresponding safety rule and follow-up item.

4 Secondary observations

Low-load integrator windup (expected, mild). This firmware predates the planned minimum-load integrator hold. During the 30–41 % of each run spent below the share-measurement floor, the gain commands drift toward their extremes (0.90–0.99), producing load-return transients of up to $-0.85 / +0.70$ share error with 0.5–1.9 s recovery — worst in exactly the two runs that also limit-cycled. Note, however, that the limit cycle itself lives at total currents up to $\approx 1.2\text{ A}$, an order of magnitude above the planned 75 mA hold threshold: the pending fix as parameterized would *not* have suppressed it.

Controller state carries across runs. Each run’s initial MDAC pair equals the previous run’s final value (only TP0007, first after boot, starts from zero). This contaminates each run’s entry

transient and should be reset at profile entry.

Run-order drift. At the identical 6 A motor command, mean bus current during the hold climbs monotonically with session order — 1.20 A (TP0007) to 1.72 A (TP0013), +44 % (Figure 7) — consistent with battery depletion and/or motor heating across the session. Cross-run comparisons are therefore made at a slowly drifting operating point; future sweeps should interleave or randomize setpoint order.

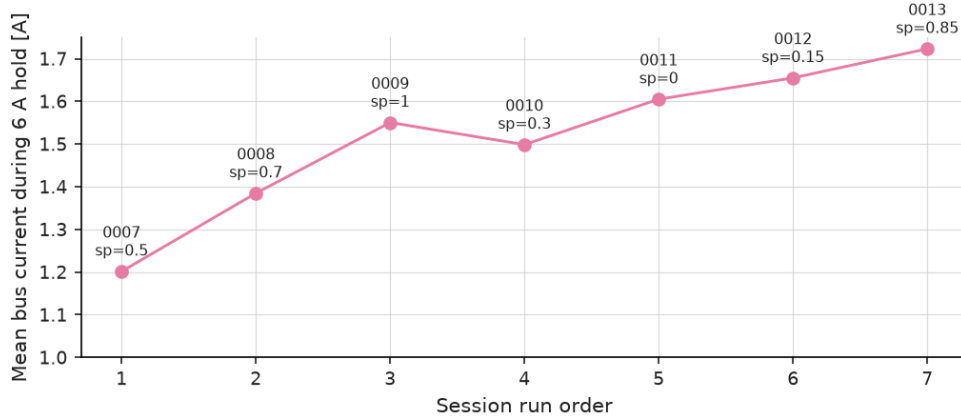


Figure 7: Session drift: mean bus current during the 6 A hold vs. run order. The +44 % monotonic climb tracks session order, not setpoint.

Comparison to the earlier runs. The five earlier captures (PS0001–PS0003 share-step profiles, TP0004–TP0005 trapezoids at $r^* = 0.5$) show clean step tracking (0.2/0.5/0.8 plateaus, settling within ≈ 1 s, no overshoot) and clean trapezoid tracking at 0.5, respectively. The new sweep reproduces that behavior at every operating point the old runs covered — there is no regression. The limit cycle appears only at asymmetric setpoints the earlier campaign never visited. (The earlier runs used less analog averaging; noise-level comparisons between the two groups are deliberately out of scope.)

5 Conclusions and recommended actions

1. **The share controller is validated across the full setpoint span at steady load:** unbiased ($< 10^{-3}$) at all seven setpoints, $\sigma \approx 0.01$ at interior points, no saturation for $r^* \in [0.15, 0.7]$.
2. **A 17–18.5 Hz minority-channel dropout limit cycle exists at asymmetric setpoints under low total current,** severe at $r^* = 0.3$ (bus to 6.5 V, 3.6 A spikes) and mild at $r^* = 0.85$. Capture 12 confirms the events are total source-feed dropouts (motor switch conducting, whole bus decaying together); droop-commanded blocking vs. source-switch SCP remains open but no longer gates the fix.
3. **Firmware follow-ups** (flagged, not yet implemented): raise or ramp-gate the minimum-load integrator hold well above 75 mA (the cycle lives up to ≈ 1.2 A); consider rate-limiting MDAC slew; reset share-controller state at profile entry; note that bus *undervoltage* is invisible under the BENCH_TEST fault set.

4. **Calibration follow-ups:** the droop authority asymmetry (FC-heavy side saturates first) and the -0.03 quantization-corrected bias at $r^* = 0.85$ both feed the pending K_{DROOP} / divider calibration round.
5. **Test-procedure follow-ups:** interleave setpoint order in future sweeps; treat $r^* \approx 0.3$ ramps as a stressor until scoped.